demand would result in losing market share, customer mind share and a leadership position; but to cater to the growth in demand would require huge investments in capacity, particularly for peak loads, and also involve additional management and operational complexities and costs.

Networks today

For the layman, the service provider network is an interconnection of routers and switches. In reality, diverse networking equipment is used in a service provider network. In addition to routers and switches, there are traffic load balancers, CG-NAT devices, content caching/serving devices (CDN), VPN boxes, a variety of security equipment starting with firewalls to intrusion detection and prevention devices to more complex threat management equipment, and many more. Figure 1 provides a conceptual illustration showcasing the diversity of equipment and their interconnections.

The service provider needs to buy the best-of-breed equipment from a variety of device vendors. Typically, each device is configured manually via Command Line Interface (CLI) by network experts trained for a particular category of devices. The way the devices are interconnected is static and limits the service flexibility that can be provided. Adding new devices or services into the mix is a daunting task.

Factors behind the rise of NFV

Server virtualisation: A little more than a decade ago, the server world underwent a transformation. It began when virtualisation technologies began to mature with support from CPU vendors. Instead of virtual machines (VMs) running on CPU emulators such as Bochs (*http://bochs.sourceforge.net/*), CPUs started to provide ways to run the VM's instructions natively. This boosted the performance of applications running in VMs to be nearly the same as when running natively. This enabled enterprises to consider server virtualisation as an option.

The migration from running applications on physical servers to running them within virtual machines enabled enterprises to make better use of their physical servers. By running services within VMs and orchestrating virtual machine creation/deletion and steering traffic to the new VM instances, enterprises were able to get elasticity, i.e., the ability to deal with fluctuating demands in an optimal manner. More VMs were provided to services seeing higher loads and less VMs to services with low load conditions.

Service providers too face similar problems. Load on a network is not the same at all times. Hence, they began to seriously consider leveraging virtualisation technologies to solve their network problems. If network services such as NAT or IDP/IPS could be run in VMs on server platforms, they realised they could save on hardware costs and reap the same benefits as enterprises.

I/O virtualisation: Once the ability to run a VM's CPU instructions at near native speeds was provided, CPU vendors started adding the ability to virtualise I/O. Until then, the I/O performance was constrained by the hypervisor's capacity or the host-OS's capacity. Thus, virtualisation was suited only for compute-heavy workloads. But, networking applications are I/O heavy.

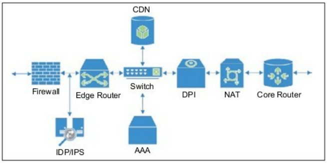

Figure 1: Today's service provider network

I/O virtualisation technologies such as SR-IOV and the ability to DMA (direct memory access) from physical devices directly into a virtual machine's address space provides a way for VMs to access the physical Ethernet ports directly, rather than via the hypervisor's device emulation. This enables the VM based applications to utilise the full bandwidth of the network ports.

Software defined networking (SDN): SDN is an emerging trend to make networks programmable. SDN is about providing applications both visibility and the ability to control how traffic flows in the network. To that end, applications are provided the required abstractions of the network, information about traffic flowing via the network and the APIs to instruct how traffic must flow.

To provide a network level view, SDN involves the separation of the data plane and control plane, moving the control plane out of the network equipment, and centralising the control plane. For interoperability and vendor independence, the data plane and control plane communicate via a standards based protocol, the most popular one being OpenFlow. It abstracts the data plane as a set of match-action rules, where N-tuple from the Ethernet-frame is looked up and the action defined by the rule is performed.

In a traditional network, each device in the network maintains its own view of the network and exchanges messages with its peers via routing protocols, so that all devices maintain the same view of the network. Configuring the control plane protocols requires special knowledge and training. When new network services are to be introduced into the network, the control plane configuration needs to be tweaked and tested, in order to ensure that there is no unintended impact on the network. This requires planning, time and effort, and hence a cost for the service providers.

In SDN, the controller takes the role of the control plane. The job of the SDN controller is to obtain information from the SDN data plane elements, build a topology, and translate the information from the network into an abstract form that can be provided to applications. Similarly, instructions received from the applications are translated into actions for each piece of equipment in the SDN network.

Coming to cloud deployments, elasticity – the ability to create or remove VMs based on load - is a key requirement for operational efficiency. This requires that traffic from the data centre gateway be steered to the new VM instance, when a VM